MTH212M PRACTICE PROBLEMS WEEK 4

Question 1. Work out the exercises left out in class.

Question 2. Fix $p \in (0,1)$. Consider the T.P. matrix

$$\begin{pmatrix} 1-p & p & 0 & 0 & 0 & \cdots \\ 1-p^2 & 0 & p^2 & 0 & 0 & \cdots \\ 1-p^3 & 0 & 0 & p^3 & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}.$$

Find the values of p such that the Markov chain on the state space $\{0, 1, \cdots\}$ becomes irreducible recurrent with period 1 for all elements. Also find $\lim_{n\to\infty} p_{00}^n$.

Question 3. Consider a Markov chain on the state space $\{0, 1, \cdots\}$ and the T.P. matrix

$$\begin{pmatrix} p_0 & 1-p_0 & 0 & 0 & 0 & \cdots \\ p_1 & 0 & 1-p_1 & 0 & 0 & \cdots \\ p_2 & 0 & 0 & 1-p_2 & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

with all $p_i \in (0,1)$. Find f_{00}^n and show that

$$\sum_{n=1}^{m+1} f_{00}^n = 1 - u_m,$$

with $u_m = \Pi_{i=0}^m (1-p_i)$. If $\sum_{i=0}^\infty p_i = \infty$, show that $\lim_{m\to\infty} u_m = 0$.

Question 4. Consider a Markov chain $\{X_n\}_{n\geq 0}$ on the state space $\{0, 1, 2, 3\}$ and the T.P. matrix

$$\begin{pmatrix} \frac{1}{6} & \frac{1}{3} & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ \frac{1}{6} & \frac{1}{3} & \frac{1}{2} & 0 \\ 0 & \frac{1}{6} & \frac{1}{3} & \frac{1}{2} \end{pmatrix}.$$

Find the communication classes. Which are transient and which are recurrent? Find all closed subsets of the state space.